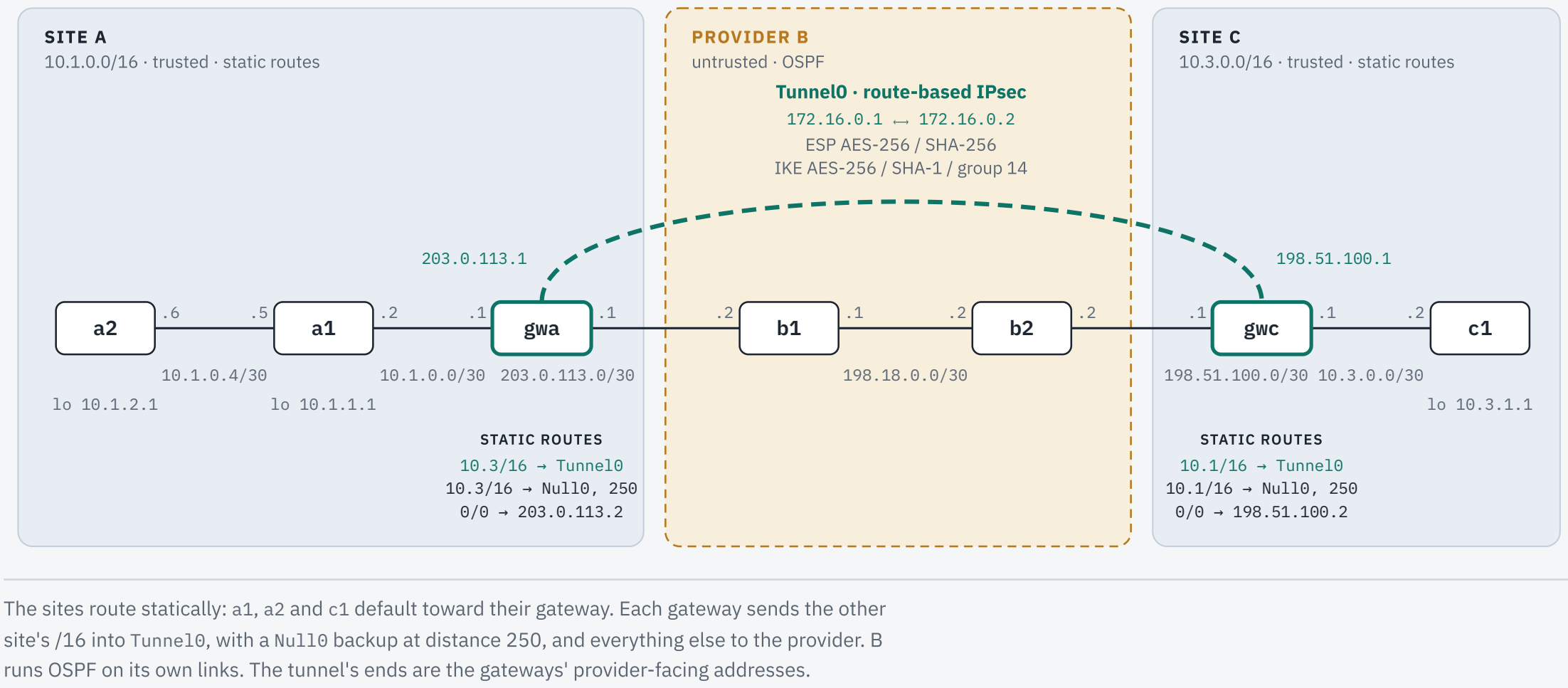
networks/ipsec-vpn

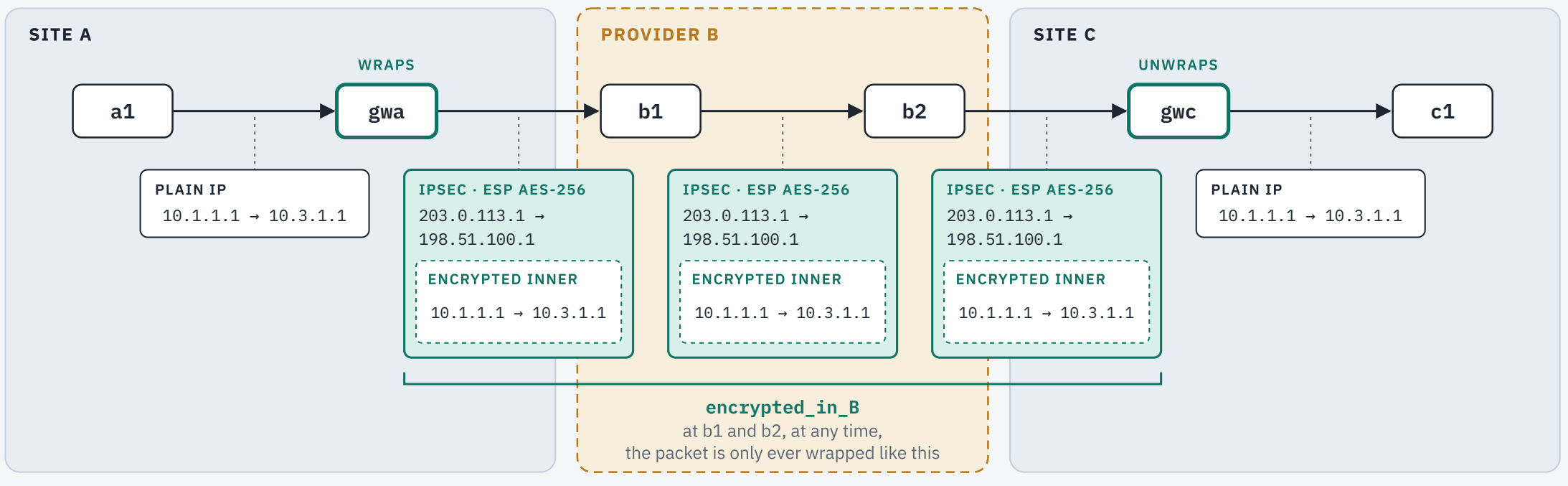
Two sites joined by a route-based IPsec tunnel across a provider they don't trust. Seven Cisco routers. The configurations are in `configs/`, and the proofs are in `Properties.lean`.

Topology



The sites route statically: `a1`, `a2` and `c1` default toward their gateway. Each gateway sends the other site's /16 into `Tunnel0`, with a `Null0` backup at distance 250, and everything else to the provider. B runs OSPF on its own links. The tunnel's ends are the gateways' provider-facing addresses.

A packet from A to C



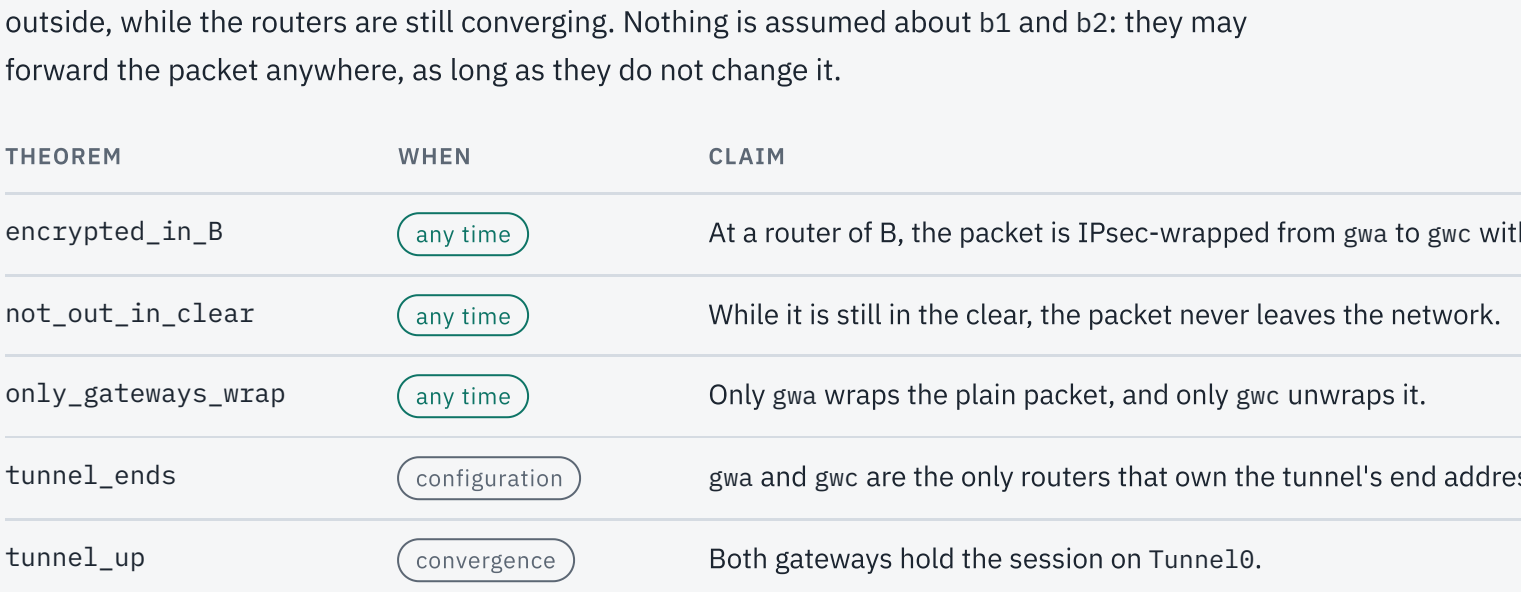
The path of `a1`'s packet to `c1` once the network has converged. `gwa` routes it into `Tunnel0`, wraps it in an outer header from 203.0.113.1 to 198.51.100.1, and sends it to `b1`. B forwards it by the outer header only. `gwc` owns 198.51.100.1 and holds the session, so it unwraps the packet and routes the inner packet to `c1`. In Lean, the packet in B is `.ipsec 203.0.113.1 198.51.100.1 proposal (.ip 10.1.1.1 10.3.1.1)`.

What is proved

These cover all 24 packets from an address of a router in A to a 10.3.0.0/16 address of a router in C. "Any time" means for every order of updates and every announcement from outside, while the routers are still converging. Nothing is assumed about `b1` and `b2`: they may forward the packet anywhere, as long as they do not change it.

THEOREM	WHEN	CLAIM
`encrypted_in_B`	any time	At a router of B, the packet is IPsec-wrapped from `gwa` to `gwc` with a proposal that encrypts.
`not_out_in_clear`	any time	While it is still in the clear, the packet never leaves the network.
`only_gateways_wrap`	any time	Only `gwa` wraps the plain packet, and only `gwc` unwraps it.
`tunnel_ends`	configuration	`gwa` and `gwc` are the only routers that own the tunnel's end addresses.
`tunnel_up`	convergence	Both gateways hold the session on `Tunnel0`.
`flows_arrive`	convergence	Every one of these packets is delivered.
`leaky_in_clear`	counterexample	Without `gwa`'s routes for 10.3.0.0/16, `a1`'s packet reaches `b1` in the clear.

Without the route into the tunnel



With `gwa`'s two routes for 10.3.0.0/16 removed, its default route sends `a1`'s packet straight to `b1`, unwrapped. So the properties above come from the gateway's configuration, not from the shape of the network.